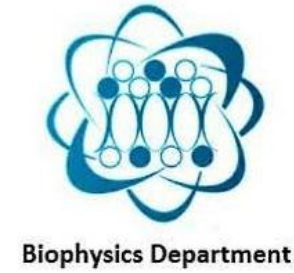
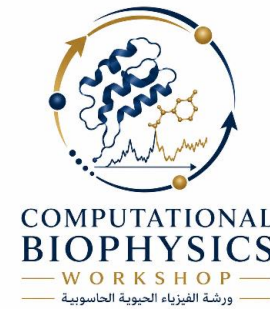


Welcome



From Sequence to Simulation: An Integrated Structural Bioinformatics Workflow – A Case Study of the GRP78–mGPX4 Interaction

Presented by:
Donia Gamal Youssef Soliman

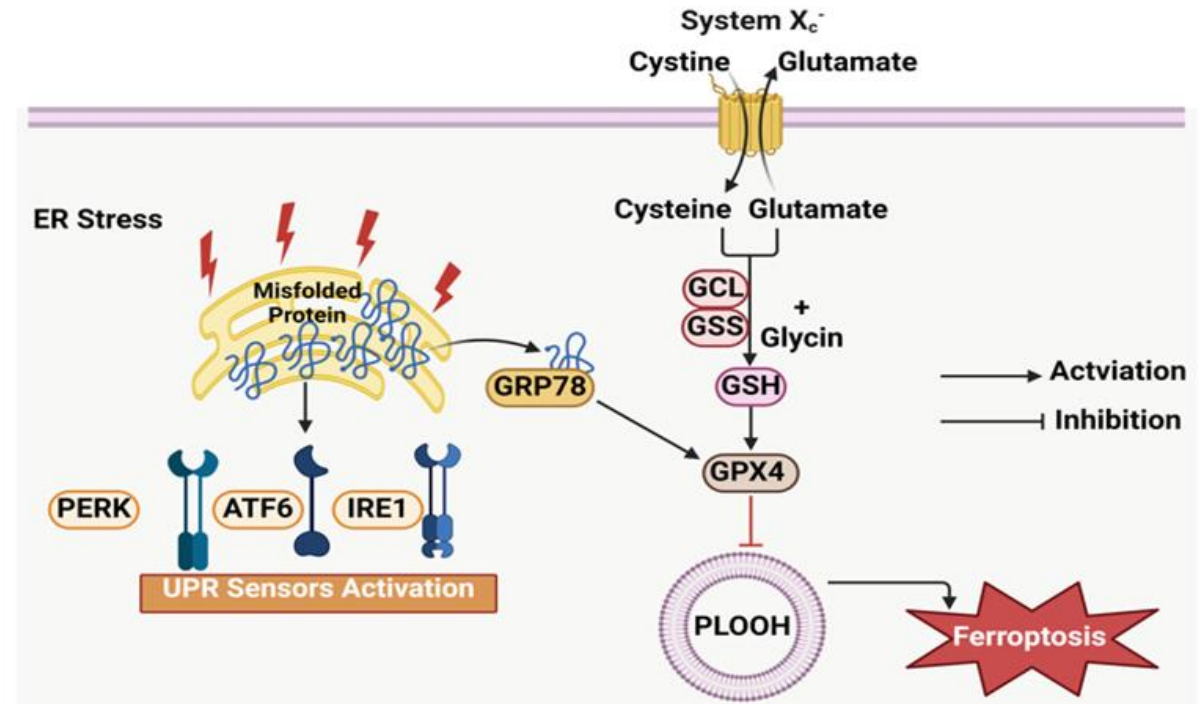
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Medical Biotechnology program
Badr University in Cairo

Contents

- **Introduction**
- **Materials & Methods**
- **Results & Discussion**
- **Conclusion**

Introduction

- **Triple-Negative Breast Cancer (TNBC):** aggressive, limited treatment options
- **Ferroptosis** and **ER stress** as key survival pathways
- **GPX4:** critical regulator of ferroptosis
- **GRP78:** master regulator of ER stress and protein folding
- **Aim:** Explore GRP78–GPX4 interaction and its role in stabilizing GPX4

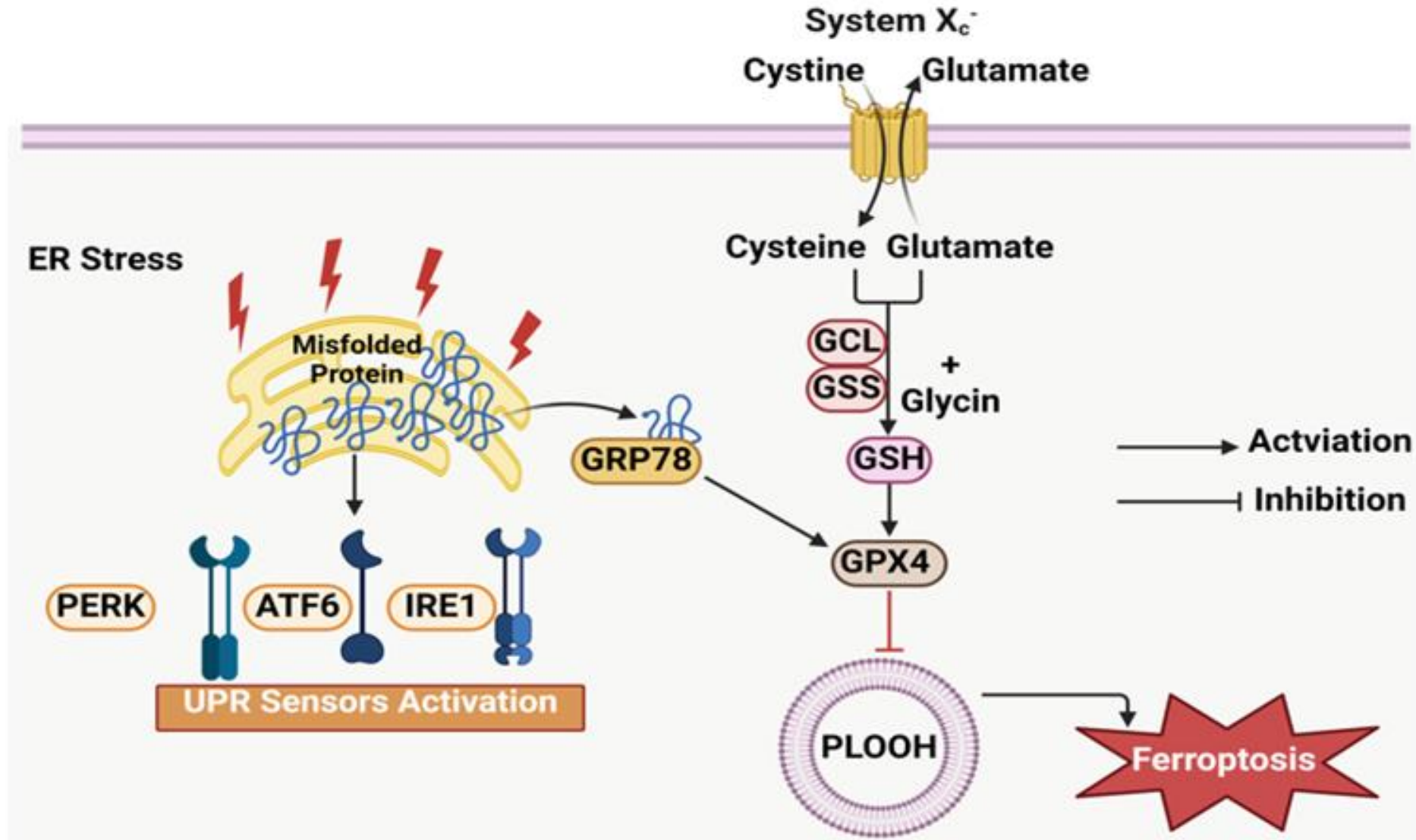


Materials & Methods

1. Peptide and Protein Sequence Analysis
2. 3D Modeling
3. Molecular Dynamics Simulations
4. Molecular Docking
5. Protein–Protein Interaction Simulations

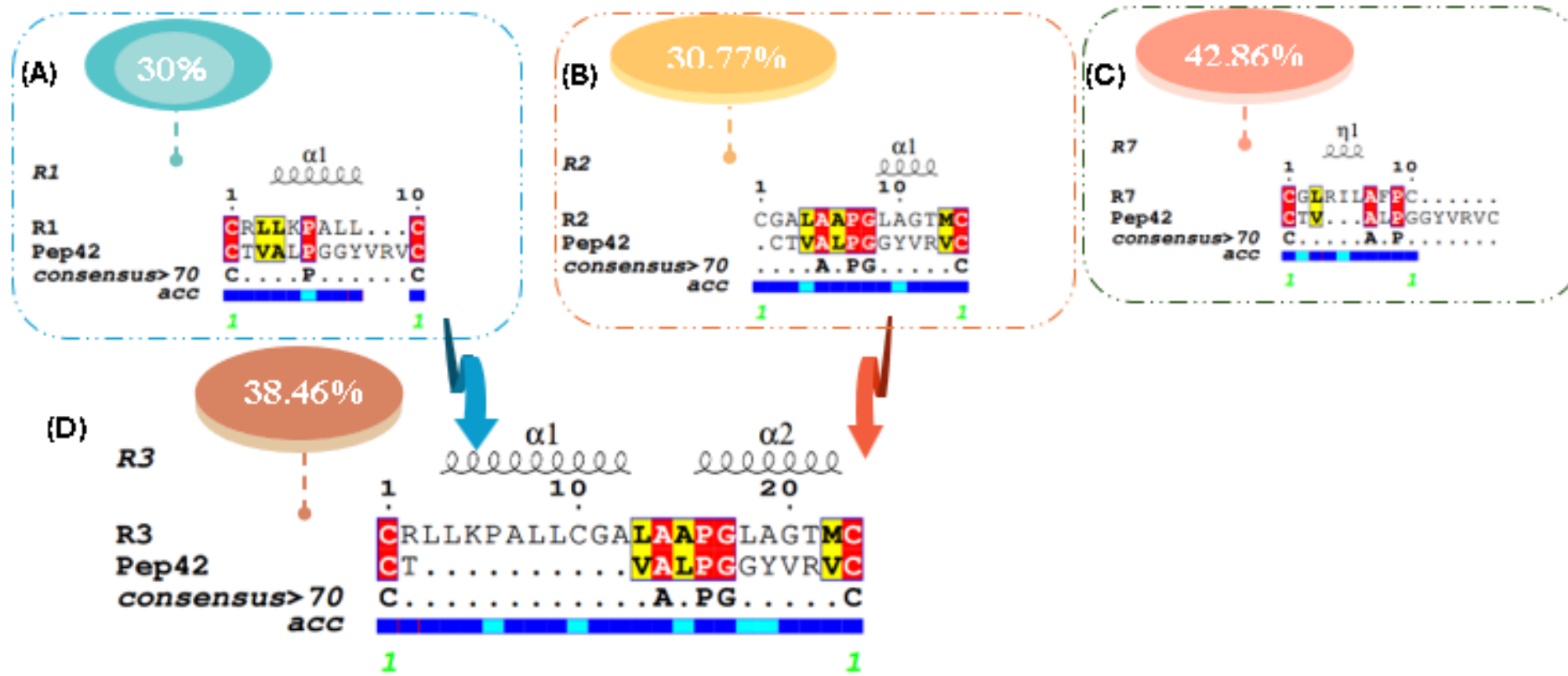
Results & Discussion

Ferroptosis and ER stress

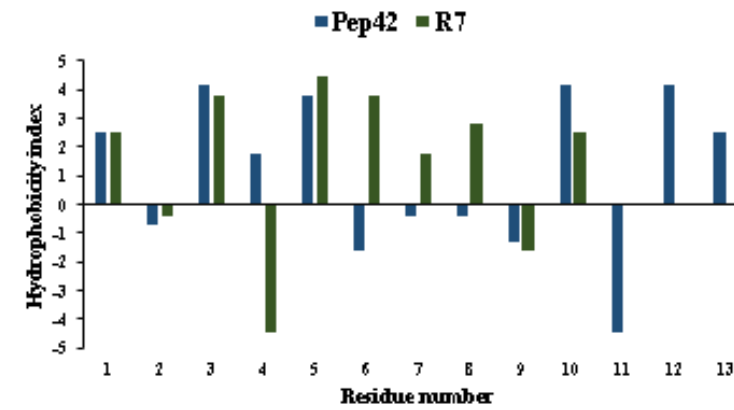
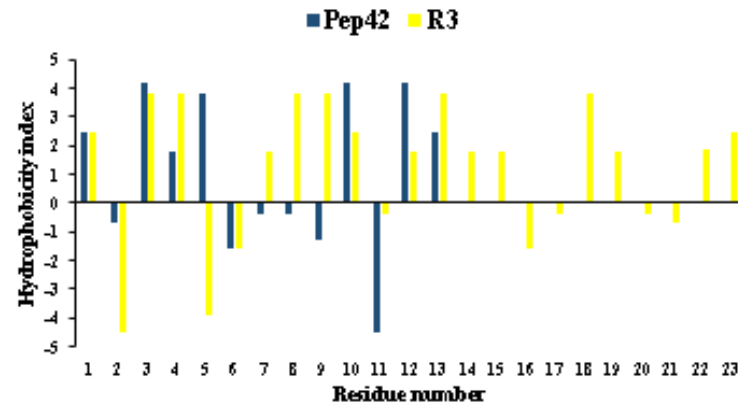
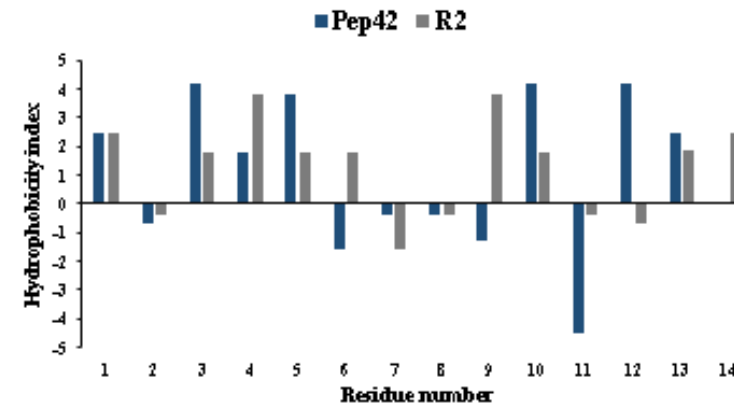
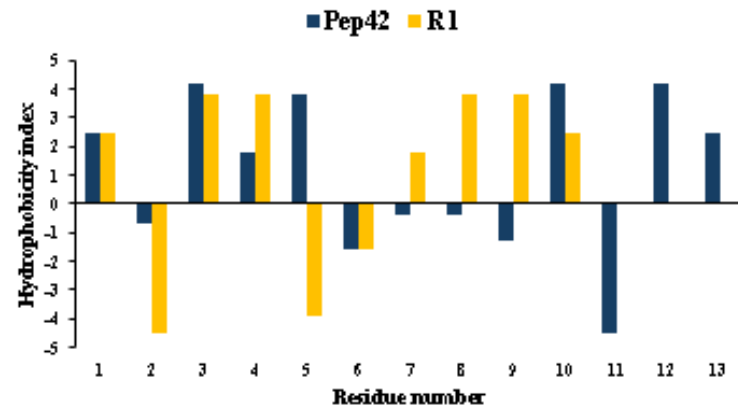


mGPX4 regions	R1	R2	R3	R4	R5	R6	R7	R8	R9
Percent Identity Matrix (%)	30%	30.77%	38.46%	22.22%	38.46%	41.67%	42.86%	23.08%	30.77%
GRAVY	1.20	1.30	1.20	-1.21	-0.35	-0.09	1.52	-0.65	-0.41
Non-polar residues content (%)	80%	92.857%	86.957 %	44.444 %	53.571%	53.333 %	90%	51.515%	59.524%
<u>Pep42</u> Grand average of hydropathicity (GRAVY): 1.10 Non-polar residue content: 84.615%									

1.2 Sequence Alignment & Visualization



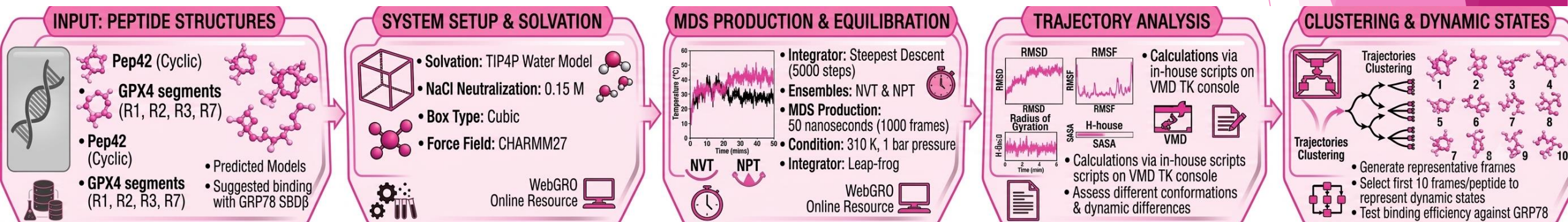
1.3 Hydrophobicity Analysis



2. 3D Modeling

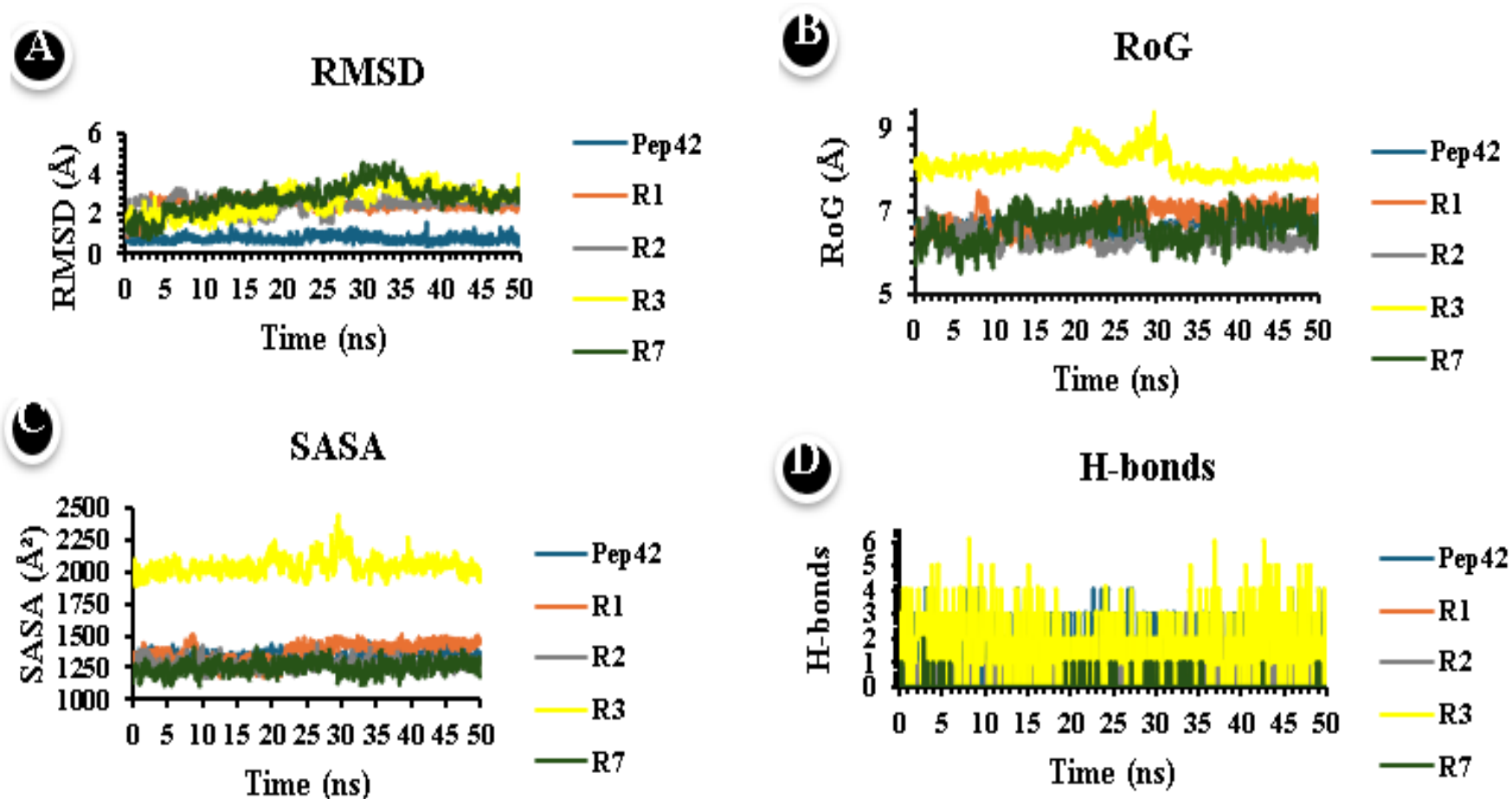
Validation Metrics	mGPX4	Pep42	R1	R2	R3	R7
Clash score	3.88	5.29	0	0	0	0
Poor rotamers Goal: <0.3%	1.23%	0%	0%	0%	0%	0%
Favored rotamers. Goal: >98%	156 (95.71%)	10 (100%)	9 (100%)	7 (100%)	15 (100%)	8 (100%)
Ramachandran outliers Goal: <0.05%	5.64%	0%	0%	0%	0%	0%
Ramachandran favored Goal: >98%	165 (84.62%)	11 (100%)	8 (100%)	12 (100%)	21 (100%)	8 (100%)
Cβ deviations >0.25Å Goal:0	2.22%	0%	0%	0%	0%	0%
Residues with Bad bonds Goal:0%	0%	0%	1 (1.3%)	0%	0%	1 (1.32%)
Residues with Bad angles Goal: <0.1%	20 (0.94%)	0%	1 (0.97%)	0%	2 (0.97%)	0%
MolProbity score	1.91	1.28	0.5	0.5	0.5	0.5

3. Molecular Dynamics Simulations



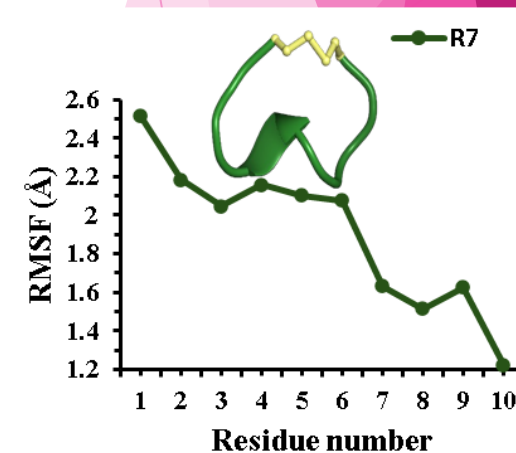
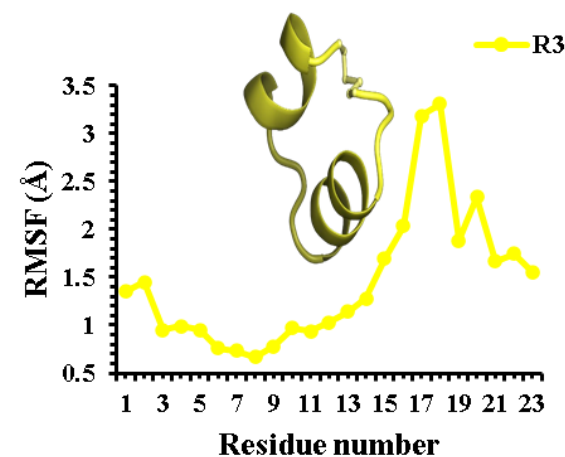
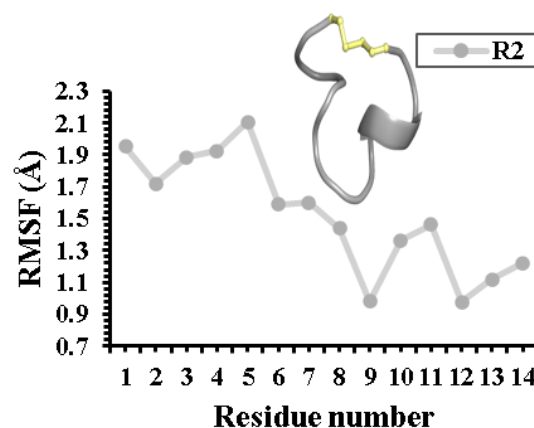
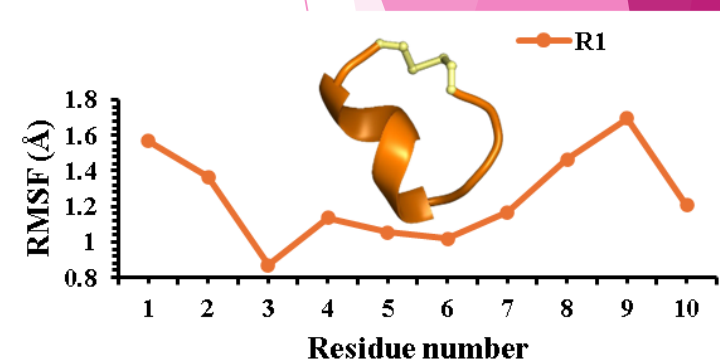
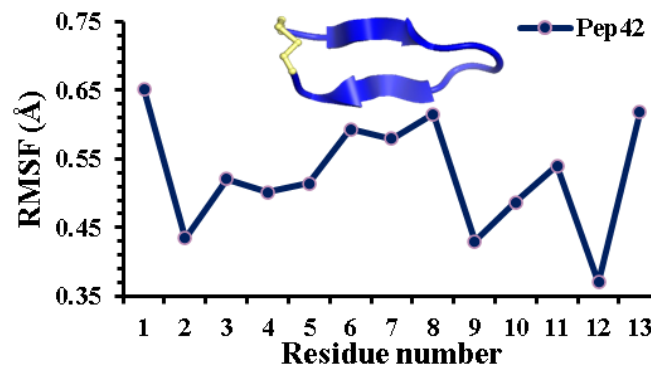
3. Molecular Dynamics Simulations

3.1 MD Stability of Pep42–GPX4 Peptides



RMSF Analysis (per residue)

- **Pep42: 0.5 ± 0.08 Å**
- **R1: 1.3 ± 0.3 Å**
- **R2: 1.5 ± 0.4 Å**
- **R3: 1.5 ± 0.7 Å**
- **R7: 1.9 ± 0.7 Å**



4. Molecular Docking

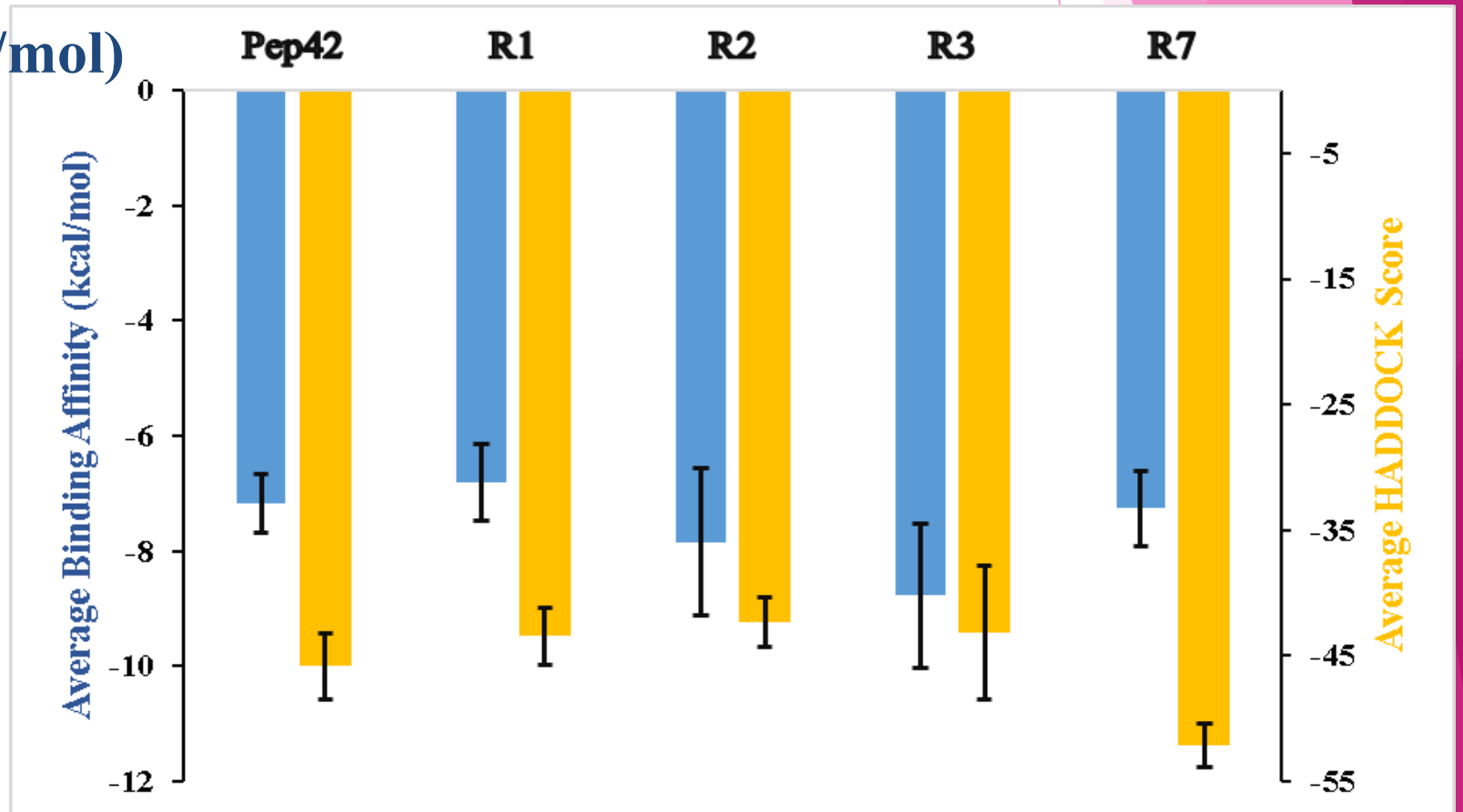
4.1 Pep42 and GPX4-derived peptides —with GRP78

■ 4.1.1 Binding Affinity (kcal/mol)

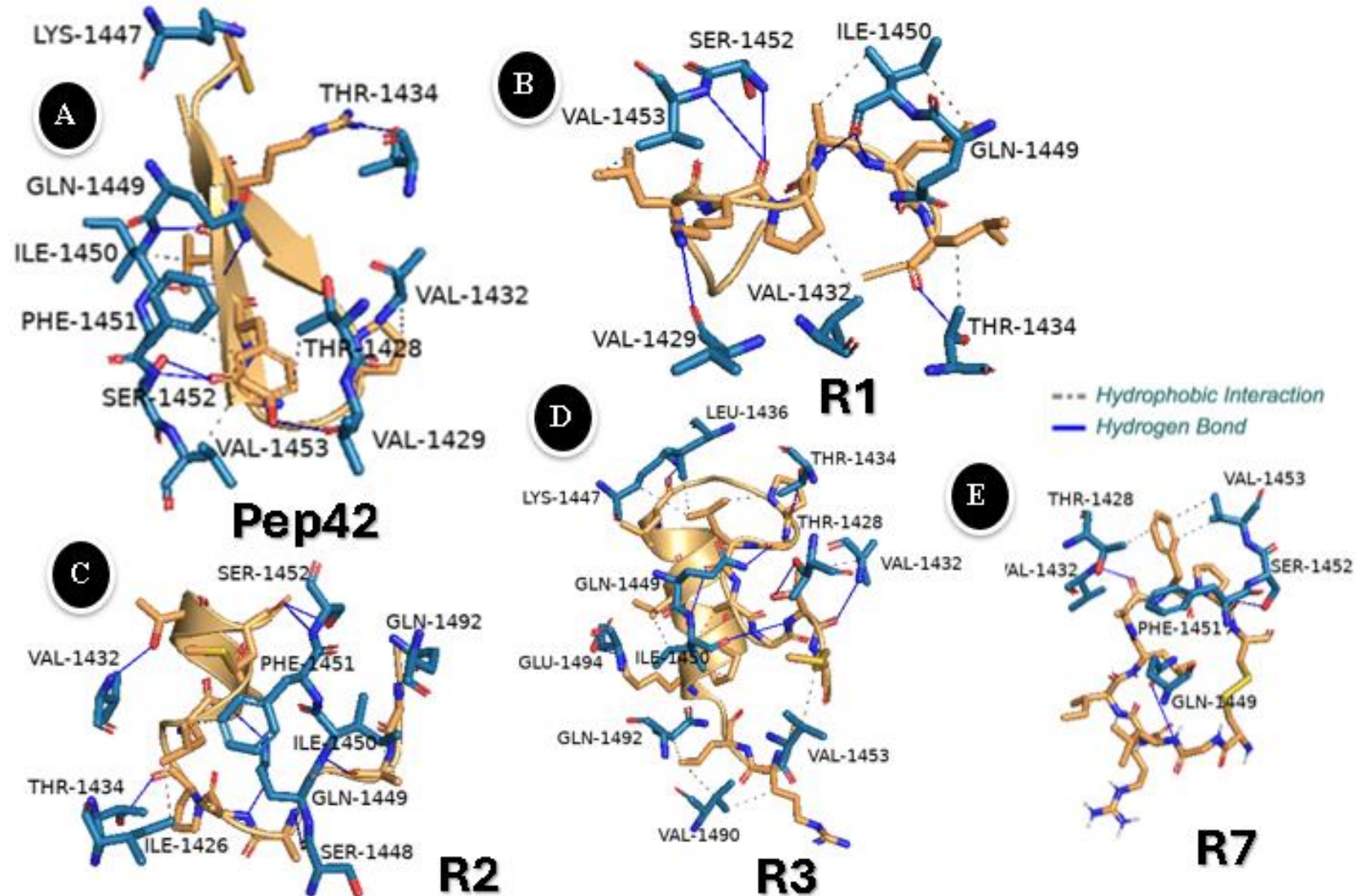
- **Pep42: -7.17 ± 0.5**
- **R1: -6.81 ± 0.7**
- **R2: -7.85 ± 1.3**
- **R3: -8.77 ± 1.3**
- **R7: -7.25 ± 0.7**

4.1.2 Docking Scores

- **Pep42: -45.82 ± 2.66**
- **R1: -43.44 ± 2.24**
- **R2: -42.34 ± 1.99**
- **R3: -43.19 ± 5.31**
- **R7: -52.12 ± 1.73**



4.1.3 Plip Analysis



4.2 Full length mGPX4–GRP78 Binding Mode

4.2.1 Binding affinities (kcal/mol)

Pep42 : -6.9 ± 0.1

Region I: -7.8 ± 0.2

Region II: -10.4 ± 0.5

Region III: -10.5 ± 0.6 (strongest)

Region VII: -7.7 ± 0.5

4.2.2 Docking scores

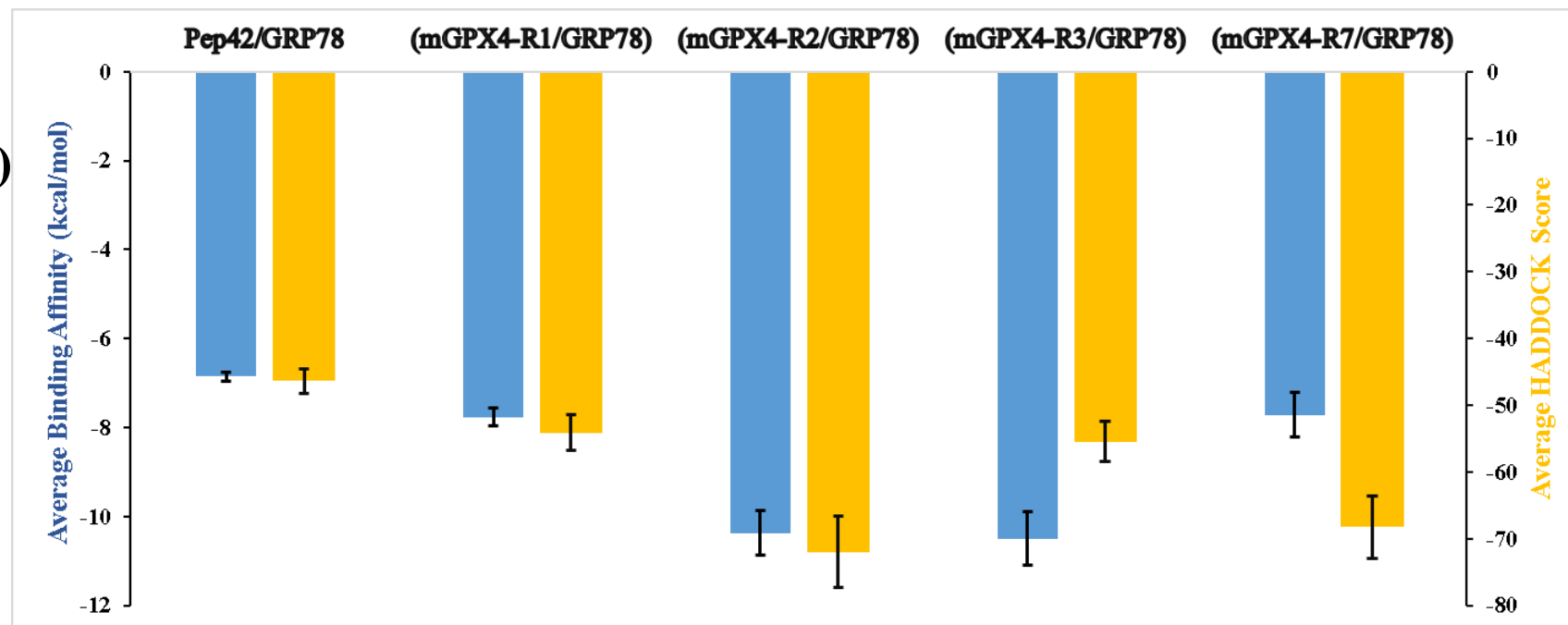
Pep42: -46.4 ± 1.8

Region I: -54.1 ± 2.6

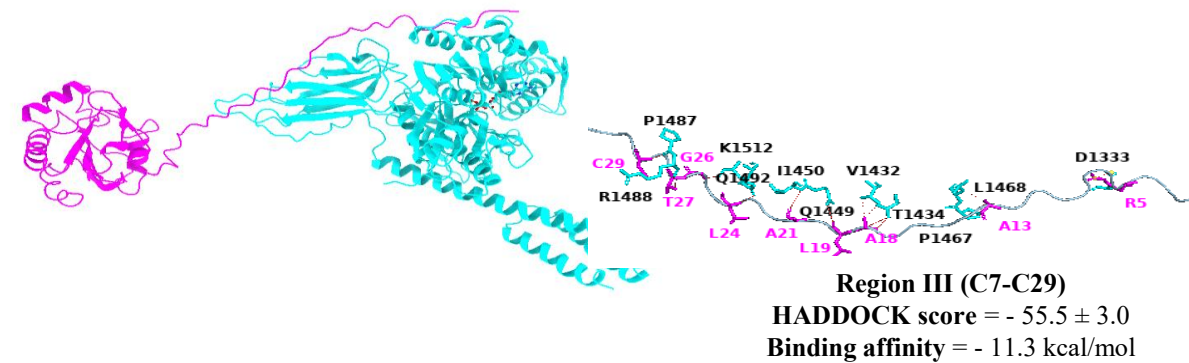
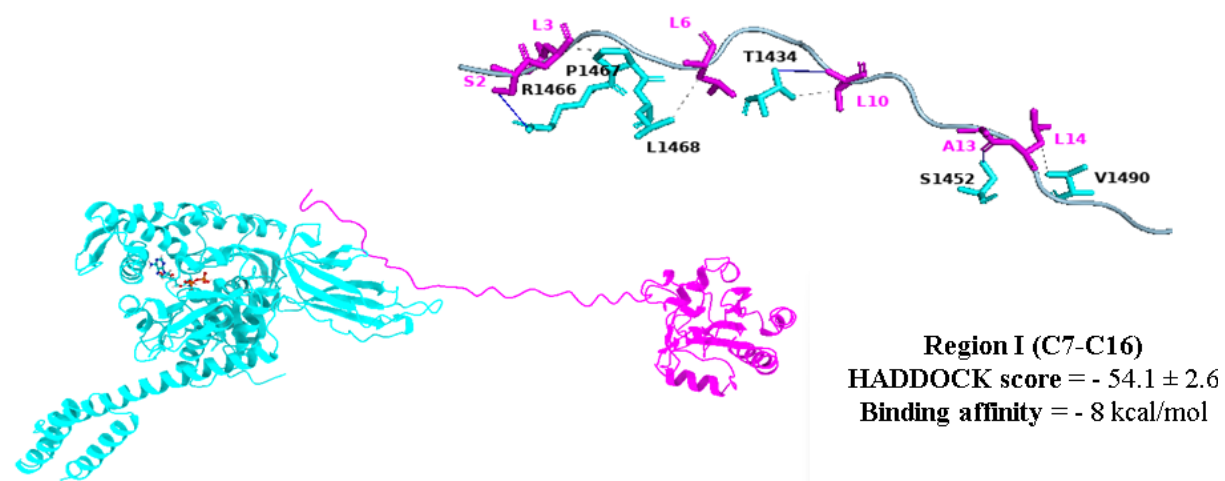
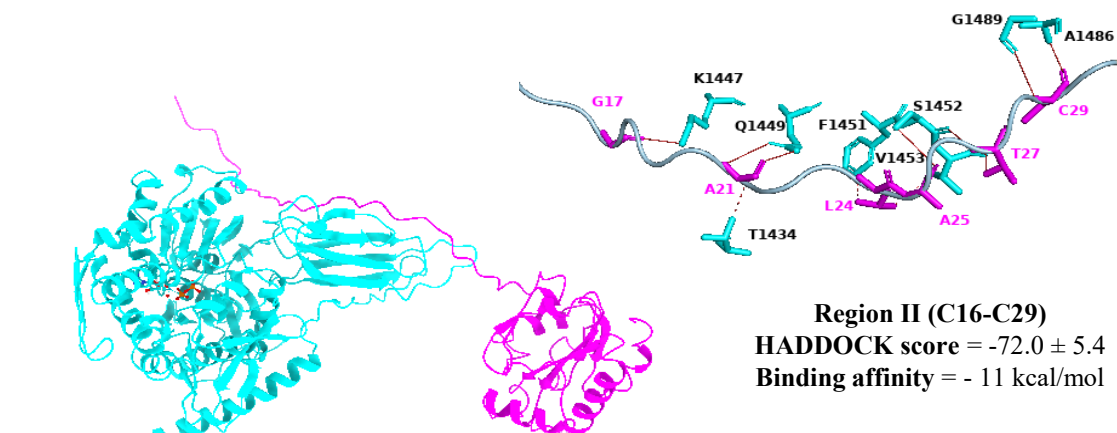
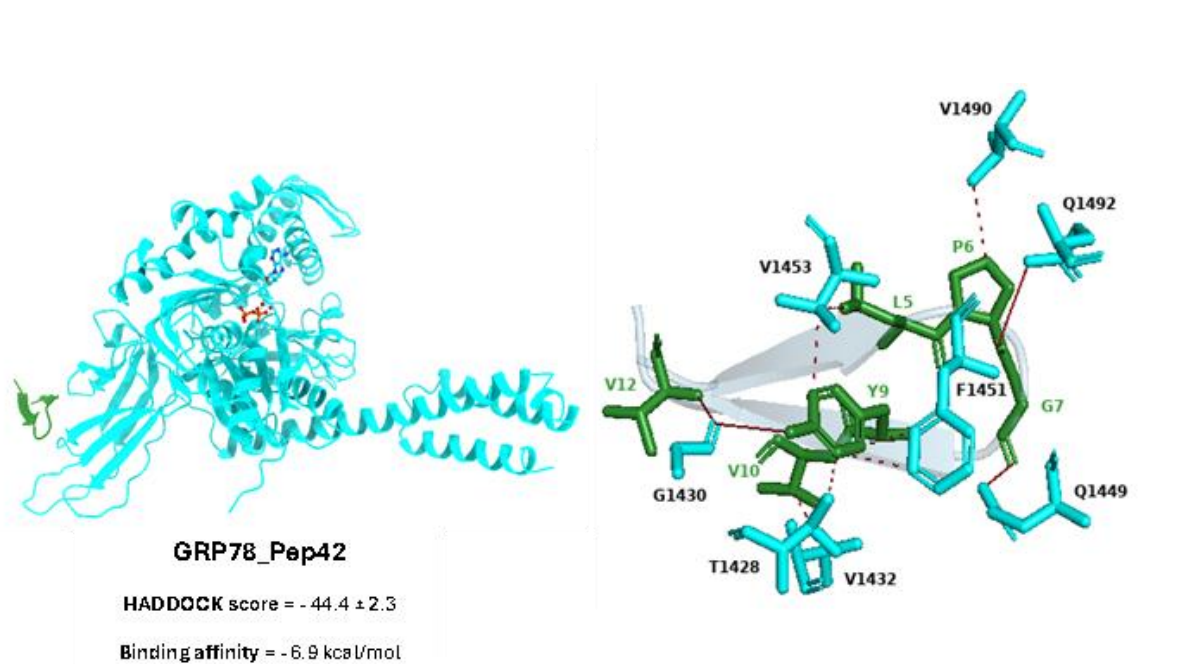
Region II: -72.0 ± 5.4

Region III: -55.5 ± 3.0

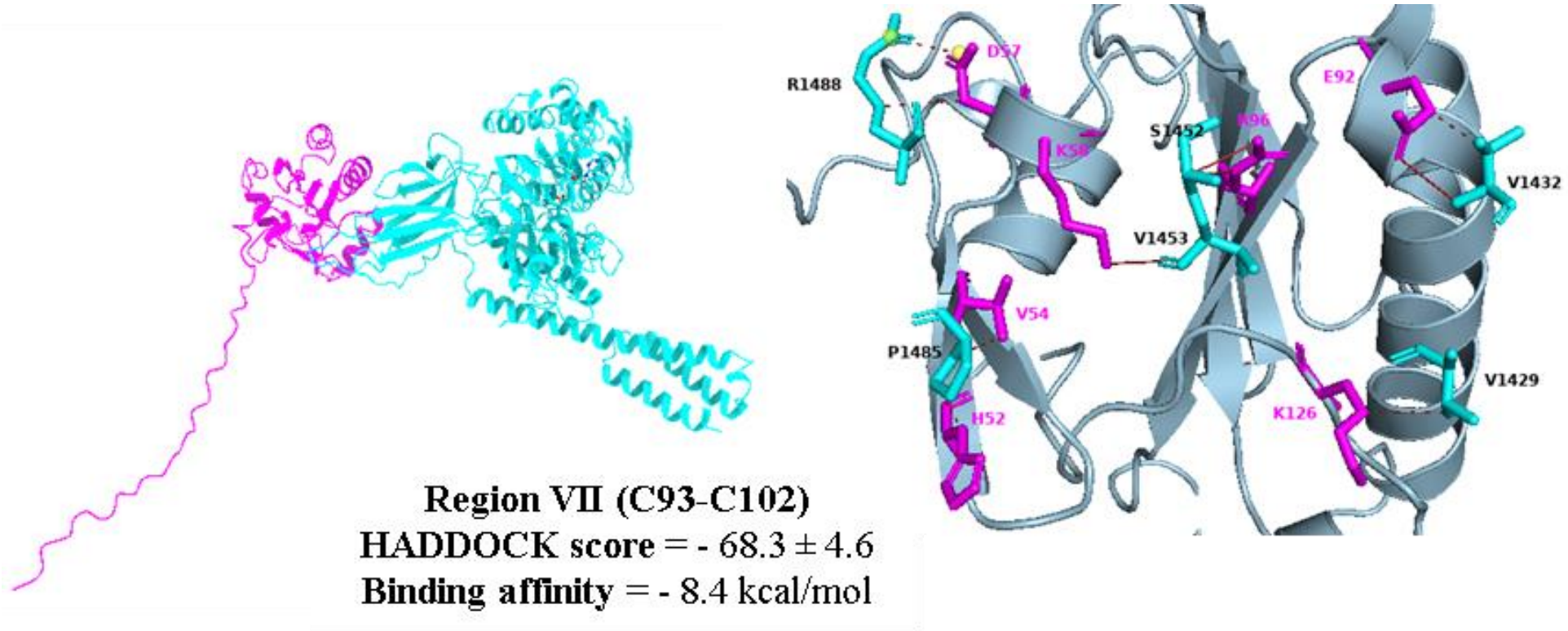
Region VII: -68.3 ± 4.6



4.2.3 Plip Analysis



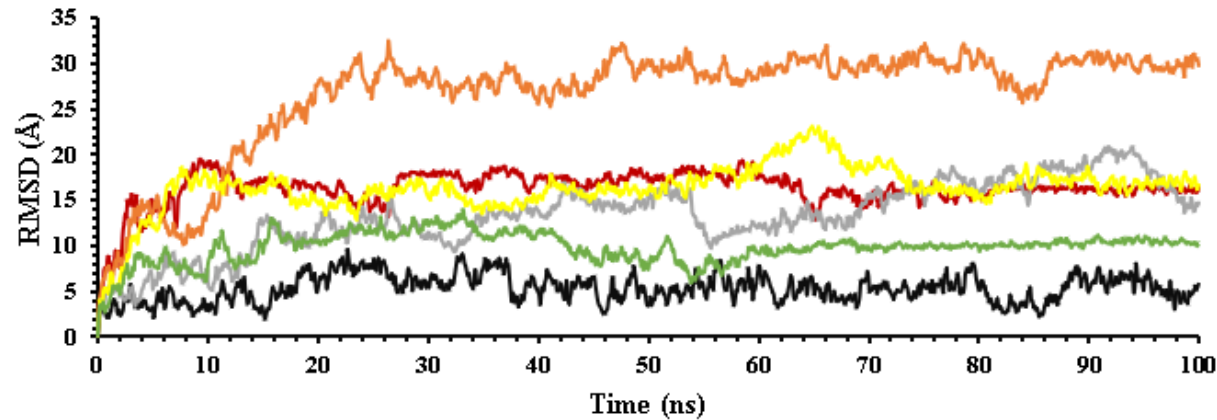
4.2.3 Plip Analysis Cont.



5. Protein–Protein Interaction Simulations

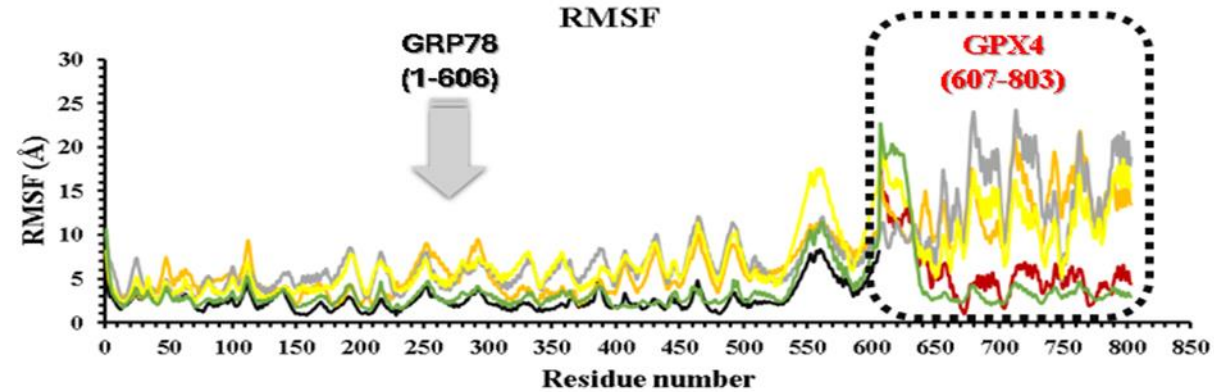
5.1 RMSD & RMSF Dynamics Analysis

RMSD



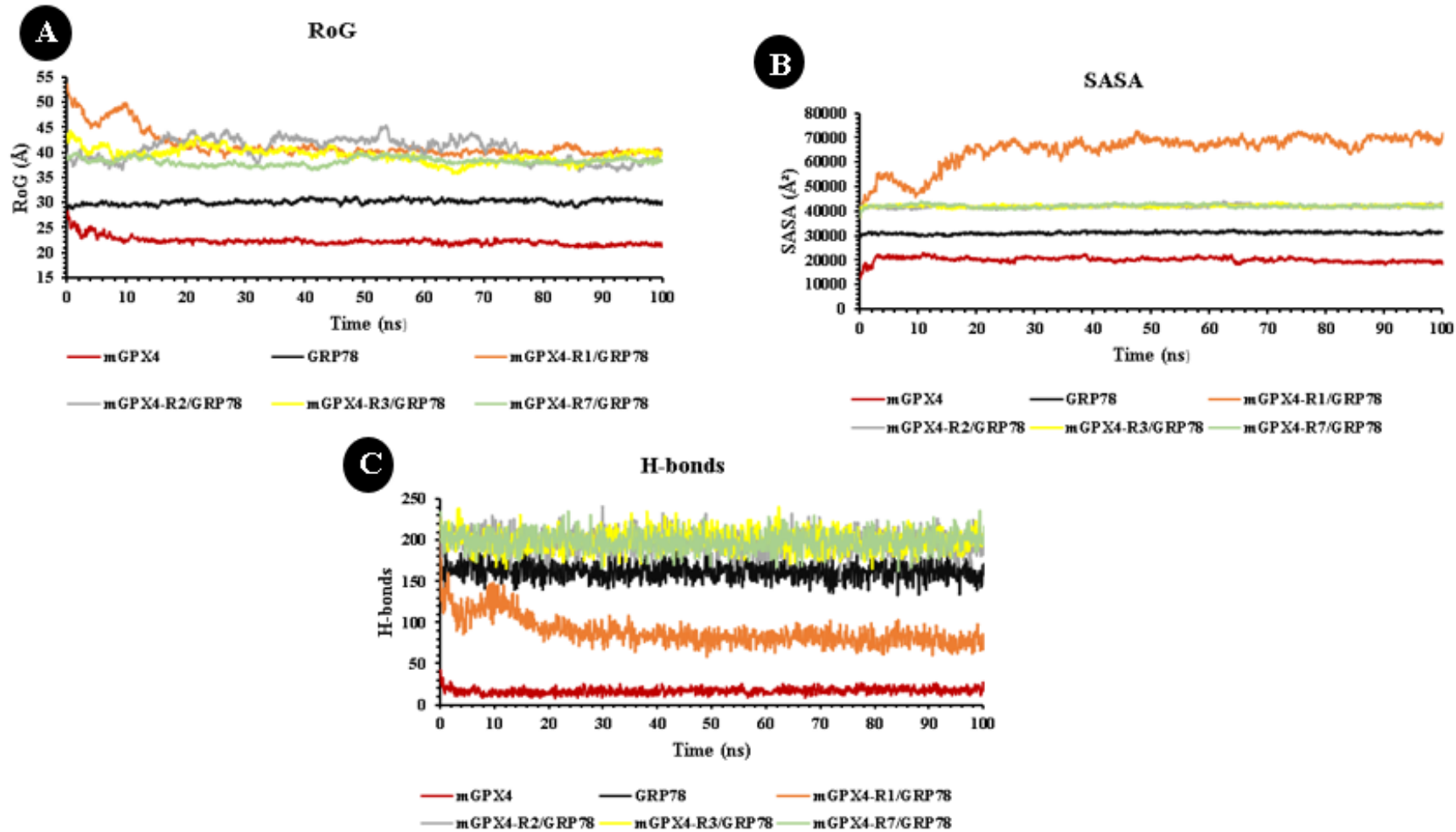
mGPX4 GRP78 mGPX4-R1/GRP78
mGPX4-R2/GRP78 mGPX4-R3/GRP78 mGPX4-R7/GRP78

RMSF



mGPX4 GRP78 mGPX4-R1/GRP78
mGPX4-R2/GRP78 mGPX4-R3/GRP78 mGPX4-R7/GRP78

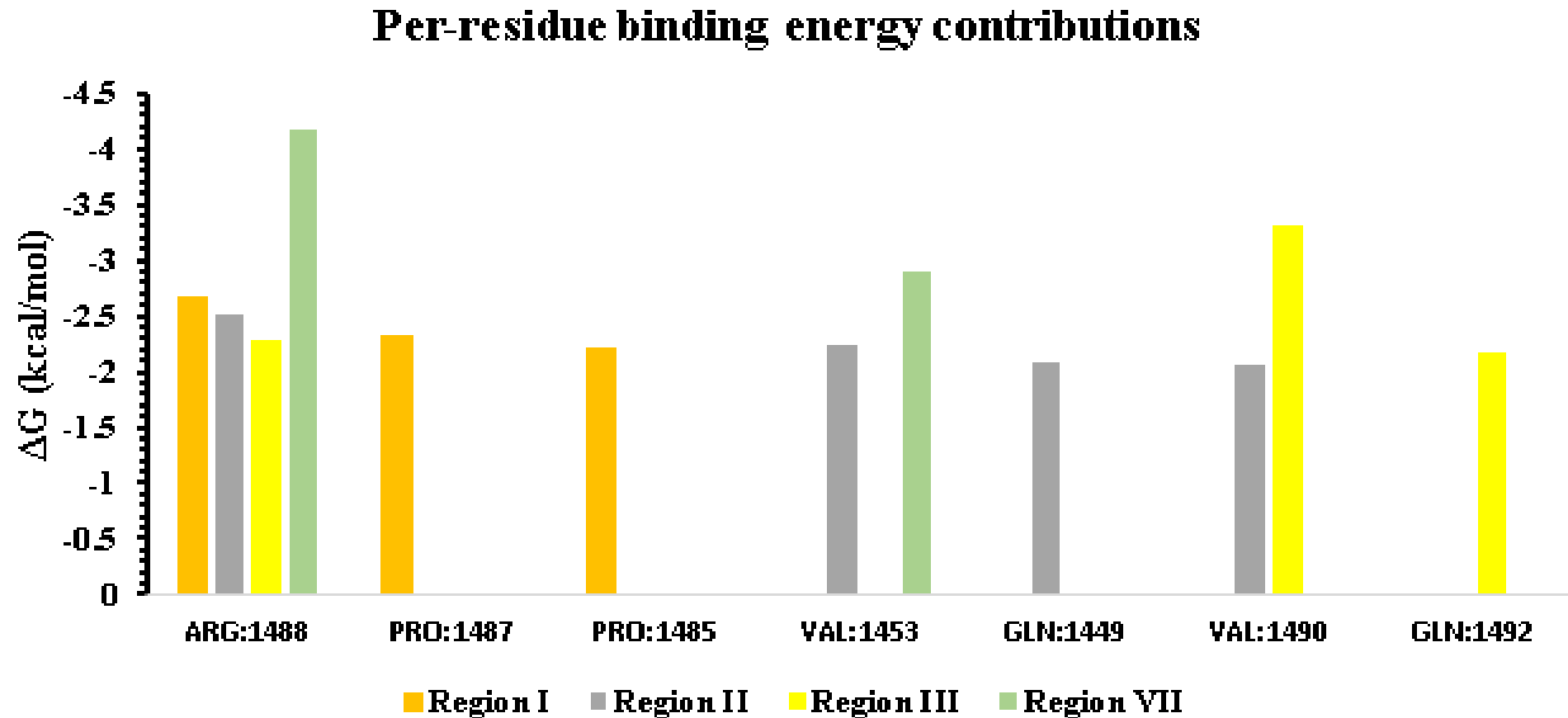
5.2 RoG, SASA, and H-Bond Dynamics Analysis



5.3 MM/GBSA Binding Free Energy Analysis

Energy Component (kcal/mol)	Complex1 (mGPX4-R1/GRP78)	Complex2 (mGPX4-R2/GRP78)	Complex3 (mGPX4-R3/GRP78)	Complex4 (mGPX4-R7/GRP78)
ΔV_{dwaals}	-50.56	-74.25	-52.58	-45.36
ΔE_{el}	-147.19	-321.56	-149.13	-82.85
ΔE_{gb}	154.38	321.70	155	89.84
ΔE_{surf}	-6.74	-12.28	-7.28	-6.83
ΔG_{gas}	-197.74	-395.81	-201.7	-128.21
ΔG_{solv}	147.64	309.43	147.72	83.01
ΔTotal	-50.10	-86.39	-53.98	-45.20

5.3.1 MM/GBSA-calculated per-residue binding free energy contributions



Conclusion

- **Our study identified novel contact sites between GRP78 SBD β and mitochondrial GPX4**
 - **Region II: highest binding energy**
 - **Region VII: most compact and stable complex**
- **These regions are critical for GRP78–mGPX4 binding**
- **Findings provide a foundation for designing inhibitors to disrupt this interaction**
- **Could help sensitize TNBC cells to ferroptosis and improve anticancer therapies**
- **Future work: Experimental validation in TNBC models needed to confirm therapeutic significance**

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Computational characterization of GRP78 binding sites on mitochondrial GPX4: implications for targeting ferroptosis in triple-negative breast cancer

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Computational investigation of the interplay between cellular stress and ferroptosis in triple-negative breast cancer

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